

ROS2 WORKSHOP

Proposed by: *robotry - by Aryan Jagushte*

Website: www.robotry.in

Trainer Profile: [Aryan Jagushte](#)

Training Mode: Offline

Course Duration: 5 Days

Workshop Objective

This 5-day workshop is designed to introduce students to ROS2 (Robot Operating System) from the ground up. By the end of the workshop, students will have hands-on experience with real-world robotics tools, environments, and algorithms.

Workshop Structure (Day-wise Breakdown)

Day 1: Introduction and Setup

- Introduction to Robotics & ROS2
- Why ROS2? Use-cases and Industry Trends
- Ubuntu Installation
- ROS2 Humble Installation & Configuration
- ROS2 Workspace & Package Structure
- Basic Linux and ROS2 Commands

| Day 2: Programming and ROS2 Architecture |

- Python Essentials for Robotics
- Build your Workspace and create Package Creation using ament_cmake
- What are Nodes, Topics, Services, Actions
- Creating Custom Publisher and Subscriber Nodes
- Create Services
- Launch Files
- Rqt graph, ros info and other commands
- Simulation with Turtlesim and Teleoperation

| Day 3: Robot Simulation & Design |

- Design Robot in Fusion 360
- Introduction to URDF
- Rqt graph, ros info and other commands
- Simulation with Turtlesim and Teleoperation
- Simulation setup in gazebo, robot state publisher and spawning
- Rviz2 and TF2 view_frames (base_link)
- Gazebo Plugins and Odometry, diff_drive, lidar plugins
- Creating Custom World; Gazebo Classic vs GazeboSim

| Day 4: Practical Implementation – SLAM & Nav2 |

- Apply SLAM & Nav2 stack in the custom environment
- Localization and map_saver
- Navigation Parameters, costmap, behaviour tree, lifecycle nodes
- What is DDS?, ROS_DOMAIN_ID
- ROS2 Communication with Microcontroller (Arduino Uno), serial communication
- Q&A and Discussion

| Day 5: ROS2 & SLAM-Nav2 |

- Lidar sensor and ROS2: rplidar/ydlidar integration
- Mapping using Lidar sensor
- Testing communication between motors, driver and controller using ros2 teleop
- Computer Vision: OpenCV and ROS2 introduction
- Q&A and Discussion

Workshop Deliverables

- ROS2 Training with Real-Time Application
 - Practical and Hands-on Exposure
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Quotation

Training Mode	Duration	Fees (INR)
Offline	5 Days	₹40,000

About robotry

Robotry is an initiative by **Aryan Jagushte**, a robotics engineer with a passion for making robotics education simple and impactful. Robotry has conducted numerous workshops and continues to bridge the gap between academia and real-world robotics.

Website: www.robotry.in

Email: admin@robotry.in / aryan@robotry.in

Phone: +91-8329574729